

Math 105LA Final Project

Due Fri 12/15, 11pm

Consider the differential equation

$$\frac{d^2 u(x)}{dx^2} = f(x), \quad x \in [0, \pi], \quad f(x) = \sin(x), \quad u(0) = u(\pi) = 0. \quad (1)$$

We develop a numerical method for solving this problem by using finite difference approximations. First we discretize the region $[0, \pi]$ by dividing it into $n+1$ parts of equal length $h = \pi/(n+1)$. The points $x_i = 0 + ih$ for $i = 0, 1, \dots, n+1$ are equally spaced point from $x_0 = 0$ to $x_{n+1} = \pi$.

Let $u_i := u(x_i)$ and $f_i := f(x_i)$.

Since $\frac{d^2 u}{dx^2}(x) = \lim_{h \rightarrow 0} \frac{\frac{du}{dx}(x + \frac{h}{2}) - \frac{du}{dx}(x - \frac{h}{2})}{h}$, $\frac{du}{dx}(x + \frac{h}{2}) = \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h}$ and $\frac{du}{dx}(x - \frac{h}{2}) = \lim_{h \rightarrow 0} \frac{u(x) - u(x-h)}{h}$, we have

$$\frac{d^2 u}{dx^2}(x) \approx \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}.$$

The finite difference approximation for the differential equation is :

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} = f_i, \quad \text{for } i = 1, 2, \dots, n \quad (2)$$

Since $u_0 = u(0) = 0$ and $u_{n+1} = u(x_{n+1}) = u(\pi) = 0$ we need not solve the differential equation at x_0 and x_{n+1} but need to simply use the boundary values of $u(x)$ given to us. Now the differential equation along with its boundary

conditions reduces to

$$\begin{aligned}
u_0 &= 0 & at \quad i &= 0 \\
\frac{-2u_1 + u_2}{h^2} &= f_1 & at \quad i &= 1 \\
\frac{u_1 - 2u_2 + u_3}{h^2} &= f_2 & at \quad i &= 2 \\
&\vdots & & \\
&\vdots & & \\
&\vdots & & \\
\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} &= f_i & at \quad i & \\
&\vdots & & \\
&\vdots & & \\
&\vdots & & \\
\frac{u_{n-2} - 2u_{n-1} + u_n}{h^2} &= f_i & at \quad i &= n-1 \\
\frac{u_{n-1} - 2u_n}{h^2} &= f_n & at \quad i &= n \\
u_{n+1} &= 0 & at \quad i &= n+1
\end{aligned} \tag{3}$$

This can be re-written as the linear system

$$\begin{bmatrix} -2 & 1 & 0 & \cdot & \cdot & 0 \\ 1 & -2 & 1 & 0 & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 0 & 1 & -2 & 1 \\ 0 & \cdot & \cdot & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ \cdot \\ u_n \end{bmatrix} = h^2 \begin{bmatrix} f_1 \\ f_2 \\ \cdot \\ \cdot \\ \cdot \\ f_n \end{bmatrix} \tag{4}$$

Now if we define $\mathbf{u} = [u_1, \dots, u_n]^T$, $\mathbf{f} = h^2[f_1, \dots, f_n]^T$ and the $n \times n$ matrix above to be A the above system of equations is written as $A\mathbf{u} = \mathbf{f}$. The Jacobi method for this problem can be written as :

$$\mathbf{u}^{(k+1)} = -D^{-1}(L + U)\mathbf{u}^{(k)} + D^{-1}\mathbf{f} \tag{5}$$

a) Show that the exact solution to system of equations $A\mathbf{u} = \mathbf{f}$ satisfies the equation

$$\mathbf{u}_{exact} = -D^{-1}(L + U)\mathbf{u}_{exact} + D^{-1}\mathbf{f}. \tag{6}$$

Thus $\mathbf{u}_{exact}$ is a fixed point for the iteration.

b) Now show that the error $\mathbf{e}^{(k+1)} := \mathbf{u}^{(k+1)} - \mathbf{u}_{exact}$ satisfies the equation

$$\mathbf{e}^{(k+1)} = -D^{-1}(L + U)\mathbf{e}^k. \tag{7}$$

c) Show that if $T_J := -D^{-1}(L + U)$ then :

$$\| \mathbf{e}^{(k+1)} \|_2 \rightarrow 0 \text{ as } k \rightarrow \infty \text{ if and only if } \rho(T_J) < 1, \quad (8)$$

where $\rho(T_J)$ is the spectral radius of the matrix T_J . Prove this result, don't just quote the theorem we did in class. I want you to prove this result using the techniques we developed in class.

d) Implement the (symmetric) power method for computing the largest eigenvalue of T_J and show that the norm of the error will decay to zero as $k \rightarrow \infty$ for the Jacobi method.

Note : The eigenvalues of course depend on the size of the system n . Further, the number of iterations to achieve a specific error tolerance depends on the size of the system n . So repeat part (d) for 3 different n values. Plot the largest eigenvalue against n .

e) Repeat part(d) for the Gauss-Seidel and SOR methods. (For SOR method,

use $\omega = \frac{2}{1 + \sqrt{1 - [\rho(T_J)]^2}}$ where $\rho(T_J)$ is approximated by the result in part

(d))

f) Which method converges faster? How does the speed of convergence depend on n ? Give an explanation of your answer.

Note:

You must write a report for this project that has the following sections:

- 1) Introduction (1-2 paragraphs)
- 2) Algorithm with a short explanation on how it works (1-2 paragraphs). This will be the algorithm used compute the eigenvalue.
- 3) Answers to all the theoretical questions of the project.

A working code alone will not get you full credit. The aim is to demonstrate your ability to design and implement an algorithm for a problem using the materials in this class.